

What I need you to measure

Updated 2026-07-31. For Mridul.

This began as a blank measurement list. It now combines your readings with datasheet values and published component envelopes. Values marked **VERIFY** are the only ones that still need the owned part.

An online/specification pass has now prefilled most of the original list. The evidence and source links are in MEASUREMENT_RESEARCH.md. Do not re-measure catalogue dimensions unless the item is marked **VERIFY** below.

M5, M11b, and M21 are now measured. No additional immediate caliper reading is required. The remaining power-fit check is selecting the smallest cable-only slot using the pigtail coupon. M25 is checked later after the driver wires are installed.

Motor identification: 28BYJ-48, 5 V. The red 2601213328 marking is a production/traceability code, not a different motor model.

Each one is written twice:

[OK] **In plain words** - what to actually do, no jargon **Technical name** - the proper term, so you recognise it in datasheets, shops and my messages

The prefilled answer sheet at the bottom shows what is known, where it came from, and what remains.

PART 0 - Words you'll keep seeing

Learn these six and everything below reads easily.

Word	What it actually means
Diameter	The width straight across a circle, through the middle. Not around it.
Bore	A hole. "Bore diameter" = how wide the hole is.
Pitch	The gap from the centre of one thing to the centre of the next - holes, pins, teeth. Centre-to-centre, never edge-to-edge.
Boss	A raised bump or collar sticking out of a surface.
Ear / lug / tab	A flat sticking-out tab with a screw hole in it.
Flat-to-flat	The width across the two flat sides of a rod that isn't fully round.

Rule of thumb: anything called "diameter" or "width" is measured with the **big jaws**. Anything called "bore" or "hole" is measured with the **small pointy jaws** on top.

PART 0.5 - Using the calipers (30 seconds)

```
+-- BIG jaws: measure the OUTSIDE of things
?
---+ +-----
| | [ 12.47 ] <- screen
---+ +-----
?
+-- small pointy jaws on TOP: measure the INSIDE of holes
```

- **mm/inch button** - make sure it says **mm**. If a big object reads 0.5 or 1.3, you're in inches.
- **ZERO button** - close the jaws fully, press ZERO, *every session*. Otherwise every reading is

off by the same amount.

- **Close the jaws gently.** Plastic squashes and you'll read too small. Stop when it stops sliding.
- **Write 2 decimal places.** "12" is useless. "12.47" is what I need.

PART 1 - THE MOTOR * MOST IMPORTANT

The silver can with the blue plastic base and 5 coloured wires.

Why this one matters most: the motor's height sets how high the base plate sits -> which sets how high the cam sits -> which sets how long the linkages are. Get the height wrong by 2mm and **all six braille dots stick up permanently** -

PART 2 - THE POSITION SENSOR

The small **blue board** with a tiny black 3-legged part on one edge, a blue adjustment screw, and pins marked AO DO GND VCC.

[!] *Bad news first*

The blue board will not fit inside the machine. It's roughly 15x11mm plus tall pins. The space inside the base plate is 5mm thick. There's no version where the whole board fits.

What has to happen: the little black 3-legged part on the edge *is* the actual sensor - everything else on that board is supporting circuitry. You'll need to **unsolder that black part** and mount just it, running three wires back.

Three solder joints. Fiddly, but genuinely the only option: it has to sit within a couple of millimetres of a magnet on the spinning disc, and the motor body fills everything else nearby.

You do NOT need to do this for the demo. On a breadboard the blue board works fine as-is.

M11 - the black 3-legged part

You can measure it while it's still soldered on.

```
+-----+ ?  
| flat | | HEIGHT (M11c)  
| face | |  
+-----+ ?  
| | | <- 3 legs  
<----->  
WIDTH (M11a)
```

from the side: <-> THICKNESS (M11b)

[OK] **M11a - How wide is the black part across its flat face?** Sensor body width *(guessed: 4.1mm)*

[OK] **M11b - How thick is it front-to-back? 1.6mm measured.** The CAD recess is also exactly 1.6mm, so it passes geometrically but has zero print tolerance. Dry-fit and clean or lightly sand the pocket before gluing; do not force the sensor. Sensor body thickness: **1.6mm measured**

[OK] **M11c - How tall is just the black block?** Not counting the legs. Sensor body height *(guessed: 3.1mm)*

[OK] **M11d - How far apart are the legs?** Middle of one leg to middle of the next. Lead pitch *(guessed: 1.27mm)*

PART 3 - THE POWER SOCKET

The 2026-07-31 photo shows a **female inline DC barrel pigtail jack**: the black cylindrical socket with red and black wires. The adapter's round male plug goes into its front.

This is the selected connector for the current pod revision. Its body remains outside the enclosure; only the red/black cable pair passes through the lid. The cable must be tied to the internal strain-relief post before it is soldered, so plug insertion/removal does not load the lid or ESP32.

Print `pigtail_slot_coupon.gcode`, then try the 5x3, 6x4, and 7x5 mm rounded slots in that order. Use the smallest slot that passes both insulated leads without scraping, pinching, whitening, or cutting them. The clipped corner marks the 5x3 end; the middle 6x4 slot is the current CAD nominal. The barrel body itself never passes through the slot.

[OK] M17 - POLARITY VERIFIED 2026-08-01

The owned pigtail was measured: **red reaches the centre pin and is positive; black is ground**. Mark the red lead physically and do not reverse it. Repeat the continuity test only if this pigtail or the adapter is replaced; the circuit has no reverse-polarity protection.

PART 4 - THE ESP32 BOARD

The black board with the silver square shield and USB-C.

M18 - RESOLVED: 15 pins per side / 30 total

[OK] **Count the metal pins along ONE long edge.** Just count them. Pin count per row

- **15 per side** (30 total) -> you have the "30-pin" board
- **19 per side** (38 total) -> the "38-pin" board

Why this matters: the pod has two printed slots the board's pins drop into. M18 confirms a 30-pin board. M21 is now measured directly from this exact board at **25.6mm**.

[OK] **M19 - How long is the board?** The long way. Board length *(guessed: 51.5mm)*

[OK] **M20 - How wide is the board?** The short way. Board width *(guessed: 28.0mm)*

[OK] **M21 - How far apart are the two rows of pins?** Middle of the left row to middle of the right row. Header row pitch: **25.6mm measured; CAD updated** *Easier: measure from the **outer edge** of the left pins to the **outer edge** of the right pins and tell me that instead.*

[OK] **M22 - The silver USB socket on the board:** how wide (a), how tall (b), and how far does it stand up off the board surface (c)? USB connector width / height / height above PCB

[OK] **M23 - Your USB cable's plug:** not the metal bit - the **plastic body behind it**, the part your fingers hold. How wide (a) and how tall (b)? Connector overmould width and height *Why: the socket sits ~2mm inside the pod, so the plug body has to pass through the hole in the wall too. The old hole was 10x7mm and would probably have blocked your cable. The service opening is now fixed at 14x9mm. M23 is optional unless your cable body is larger than that.*

PART 5 - THE MOTOR DRIVER (blue board, 4 red LEDs)

[OK] **M24 - How long and how wide is the blue board?** PCB length x width

[OK] **M25 - How tall is it once the wires are soldered flat against it?** Assembled stack height *Measure **after** soldering, or press the wires flat and estimate. Do **not** measure it with the plug-in jumper wires standing upright - that makes it ~20mm and the space is 16mm. That's exactly why the wires must be soldered flat.*

[OK] **M26 - Which part on the board stands up tallest, and how tall is it** from the board surface? Tallest component height

PART 6 - Quick confirmations (10 seconds each)

[OK] **M27 - One of your silver disc magnets: how wide and how thick?** Magnet diameter x thickness *(expect 8mm x 1mm)*

[OK] **M28 - One tiny black push-button: how wide is the square body, and how tall is it including the round nub on top?** Tactile switch body and total height *(expect 6 x 6 x 5mm)*

[!] Shopping correction that came out of this work

Your BOM used to say buy a **3mm x 2mm** homing magnet.

Buy 3mm x 1mm instead.

The disc it glues into is only 2mm thick. A 2mm-deep pocket cut clean through it, punching a 3.2mm hole across three of the six cam tracks - and a linkage foot crossing that hole would drop in and jam the whole mechanism. A 1mm magnet leaves 0.8mm of material and works just as well, because it still sits flush against the underside.

Already ordered 3x2mm? They'll be useful elsewhere. Just buy 3x1mm too. ~?120 for ten.

? CURRENT MEASUREMENT RECORD

This is the consolidated owned-part/specification record. Blank catalogue-only items are no longer immediate blockers unless a first-print dry-fit shows a conflict.

```

=== MOTOR (owned 28BYJ-48, 5V; red number is a lot/traceability code) ===
M1  can diameter                = 28.1 mm    [MEASURED]
M2  mounting-face height       = 19.0 mm    [MEASURED]
M3  shaft offset from can centre = 7.5 mm     [MEASURED]
M4  shaft diameter             = 5.0 mm     [MEASURED + SPEC]
M5  shaft across flats         = 3.0 mm     [MEASURED]
M6  mounting face to shaft tip  = 9.5 mm     [MEASURED]
M7a raised collar diameter     = ~9.0 mm    [SPEC]
M7b raised collar height       = 2.0 mm     [DERIVED: 9.5 - 7.5]

```

```

M8 mounting-hole pitch          = 34.7 mm [MEASURED]
M9 mounting-hole bore          = 4.0 mm [MEASURED]
M10a mounting-ear width        = ~7.0 mm [SPEC]
M10b mounting-ear metal thickness = ~0.8 mm [SPEC]

=== POSITION SENSOR (KY-024 family, bare 49E/SS49E package) ===
M11a body width                = 4.1 nominal / 4.2 max mm [SPEC]
M11b body thickness            = 1.6 mm [MEASURED - EXACT-FIT POCKET]
M11c body height               = 3.1-3.2 mm [SPEC]
M11d straight-lead pitch       = 1.27-1.30 mm [VERIFY LEAD FORM]

=== POWER SOCKET (photo identified) ===
M12-M14 owned part            = inline female DC pigtail [CURRENT POD CHOICE]
M15 mating barrel              = likely 5.5 OD / 2.1 centre mm [CONFIRM WITH PLUG]
M16 panel thread + ring nut    = NO [PHOTO CONFIRMED]
M17 polarity                   = red -> centre pin / +5V [MEASURED 2026-08-01]

=== ESP32 BOARD (USB-C, CH340C-family, 15 pins per row) ===
M18 pin count                  = 15 per row / 30 total [MEASURED]
M19 board length               = likely 51.5-52.0 mm [MATCHING LISTINGS]
M20 board width                = likely 28.0-28.5 mm [MATCHING LISTINGS]
M21 header-row centre spacing = 25.6 mm [MEASURED - CAD UPDATED]
M22 USB-C interface / shell    = ~8.3x2.5 / ~9x3.2 mm [SPEC/LIKELY]
M23 cable body                 = NOT NEEDED unless larger than 14x9 mm

=== MOTOR DRIVER (common blue four-LED ULN2003 board) ===
M24 PCB size                   = likely 35x32 mm [COMMON VARIANT]
M25 installed height, cable bent flat = likely ~12 mm [VERIFY AFTER WIRING]
M26 tallest item               = JST-XH motor plug, ~10-12 mm installed [LIKELY]

=== QUICK CONFIRMATIONS ===
M27 docking magnets            = 8x1 mm [OWNED / ALREADY IN CAD]
M28 tactile switches           = 6x6x5 mm [PURCHASE SPEC / ALREADY IN CAD]

```

What happens when you send these back

I re-derive the **whole vertical stack in one pass** - motor height -> base plate -> cam -> linkage length - rather than patching one number at a time. That's deliberate: these all depend on each other, so fixing them individually just moves the error somewhere else.

The remaining hardware decisions/checks are:

Waiting on	Why online data is not enough
Hall pocket dry-fit	M11b is 1.6mm and the recess is exactly 1.6mm, so real printed tolerance must be checked before gluing.
Pigtail cable-slot coupon	Select the smallest 5x3 / 6x4 / 7x5 opening that passes both insulated leads without damage; feed the result into power_cable_slot_w/h.
M25 later	Final height depends on how the motor cable and soldered wires are dressed during assembly.

Everything else is now measured, derived, standardized, safely enveloped, or non-critical.